Amirhossein Dalvand

Shiraz, Iran
Sharif University of Technology
dalvand.ah@gmail.com

Education

M.Sc., Pure Mathematics 2023-Present
Sharif University of Technology, Iran
Total GPA: 4/4 (19.23/20)
Ranked 2nd

B.Sc., Pure Mathematics 2018-2023
Shiraz University, Iran
Total GPA: 3.95/4 (18.42/20)
Ranked 1st

Research Interests

Category Theory · Applied Category Theory · Mathematical Biology · Synthetic Biology

Honors and Awards

- **Ranked 6th** among approximately 3,000 participants in the National Graduate Entrance Examination in Mathematics 2023
- **Silver Medal**, 44$^{\text{th}}$ Iranian Mathematics Competition for University Students, Tehran, Iran May 2022
- **Ranked 3rd** in the National Scientific Mathematics Olympiad for University Students, Tehran, Iran Nov 2021

Research Experience

- **M.Sc. Thesis, Sharif University of Technology** 2024–Present
  Thesis: "An Extension of Order Topology Convergence to Category Theory and Its Applications"
  Supervisor: Prof. Amir Daneshgar
- **Research Project: Secure Voting Protocols** 2022–2023
  Developed a secure voting protocol based on secret sharing and zero-knowledge proofs.
  Resulted in the preprint: "A Secure Voting Protocol Using Secret Sharing and Zero-Knowledge Proofs".

Selected Projects

- **ECG Signal Analysis and R-Peak Detection** 2024
  Studied and improved existing ECG R-peak detection algorithms based on visibility graphs.
  Implemented and evaluated modifications in Python.
- **Form Recognition using Morphological Image Processing** 2024
  Implemented an image classification system based on morphological filtering techniques.
  github.com/dalvand-ah/morphological-form-classifier

Academic Activities

- Seminar Talk: Existence and Uniqueness of the Haar Measure — Spring 2025
- Seminar Talk: Variational Problems in Discrete Geometry — Spring 2025
- Course Presentation: Multivariate Cointegration in Statistical Arbitrage — Fall 2024
- Organizer and Instructor, Problem-Solving Workshops for Undergraduate Mathematics Competitions, Shiraz University — Fall 2022
- Organizer of Biweekly Paper Reading Sessions for Undergraduate Mathematics Students — Summer 2022
- Organizer of the "One Step in Algebra" Seminar Series, Shiraz University — Nov 2021

Teaching Experience (Teaching Assistant, Shiraz University)

- Mathematical Analysis I — Fall 2022
- Complex Functions — Fall 2022
- Algebra — Fall 2021
- Mathematics and Its Applications in Management — Spring 2021

Volunteer Activities

- Staff Member, Mathematics Library, Shiraz University — 2021
- Member of the Mathematics Student Association, Shiraz University — 2020

Technical Skills

Programming: Python, C, MATLAB
Typesetting: LaTeX